Lam Nguyen, Ph.D.

AI Security & Privacy Engineer | Mobile Security Researcher | Applied AI / Agentic AI Systems | DevOps / Platform Engineer

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Professional Summary

Postdoctoral Researcher and former Senior DevOps/Backend Engineer with experience in **mobile security and privacy**, **Android app analysis**, **applied AI/LLM systems**, and **cloud-native DevOps automation**. Experienced in designing, implementing, and maintaining scalable systems for both research and production environments, including enterprise API gateway infrastructure serving high-volume daily traffic.

Through my research on security and privacy in mobile ecosystems, I have designed and built automated analysis pipelines capable of detecting privacy violations, sensitive data leakage, and privacy-policy inconsistencies across thousands of Android applications. These systems combine **Python, Android tooling, static/dynamic analysis, taint analysis, network traffic inspection, LLM reasoning, RAG/GraphRAG, and agentic AI** to improve scalability, automation, and accuracy beyond traditional manual analysis workflows.

I also have practical industry experience in backend engineering and DevOps, including **Go, Docker, Kubernetes, Kong API Gateway, CI/CD, Jenkins, GitLab CI/CD, Ansible, Istio, Linux, and ELK**. At Home Credit Vietnam, I worked on OpenAPI gateway automation, Kong deployment and configuration, GitOps workflows, zero-downtime canary releases, backup/disaster recovery, and security hardening for production API infrastructure.

Target Roles

AI Security & Privacy Engineer; Mobile Security Researcher; Applied AI / LLM Engineer; Agentic AI Engineer; DevSecOps Engineer; Backend / Platform Engineer.

Core Technical Skills

Programming	Python, Go, Android, PHP, Bash, Linux command line
AI / LLM Systems	RAG, GraphRAG, RAG-Fusion, Corrective RAG, few-shot learning, fine-tuning, LangChain, LangGraph, MCP, Agent-to-Agent (A2A)
Security / Privacy	Android security, mobile privacy, privacy compliance, GDPR/CCPA-regulation data-practice auditing, Data Safety inconsistency detection, static analysis, dynamic analysis, taint analysis, reverse engineering, MITM proxy-based traffic inspection
DevOps / Platform	Kubernetes, Docker, Jenkins, GitLab CI/CD, Harbor Registry, Ansible, Rancher, Istio, Kong API Gateway, GitOps, Elastic Stack, F5 firewall, AWS
Databases / Graphs	Neo4j, MongoDB, MySQL, knowledge graphs
Blockchain	Hyperledger Fabric, Hyperledger Besu
Languages	Vietnamese native; English professional proficiency

AI, Security, and Privacy Engineering Experience

Postdoctoral Researcher – Agentic AI for Mobile Security and Privacy 11/2025 – Present
Department of Theoretical and Applied Sciences (DiSTA), University of Insubria Varese, Italy

- **WearPri**: Leading the design of a pre-release privacy auditing system for Android wearable companion apps using **taint analysis, GraphRAG, RAG, and LLM-based code summarization** to estimate data-sharing non-compliance and attribute responsibility to app developers or third-party services.
- **EBPS – Evidence-Based Policy Synthesis for Android**: Designing a context-aware access-control and privacy-policy synthesis framework that uses **LLM reasoning, multi-source evidence modeling, and policy optimization** to derive enforceable privacy-policy candidates beyond static binary permission models.
- **RootFlow**: Designing a root-anchored mobile app exploration framework that builds **feature-flow graphs** from UI observations, state normalization, semantic annotation, and root-return path verification to synthesize executable goals for LLM-based mobile agents such as DroidRun.

Ph.D. Candidate – Mobile and Wearable App Security/Privacy 11/2022 – 11/2025
University of Insubria Varese, Italy

- **MetaLeak**: Built a hybrid Android app analysis system combining **static permission analysis, dynamic traffic inspection, and MITM proxy-based monitoring**; evaluated on **5,000 apps** and found that **21.9%** transmitted sensitive image metadata.
- **ALIBIS**: Developed an automated source-code risk estimation system using **static analysis, code-block extraction, RAG, and LLM-based code summarization**; achieved accuracy **0.8686**, precision **0.8902**, recall **0.881**, and F1-score **0.8854**.
- **WearLeak**: Built a privacy-compliance analysis pipeline for wearable companion apps using **dynamic traffic analysis, Data**

Safety disclosure analysis, LLMs, and GraphRAG; evaluated **711 apps**, finding **67.5%** with declared data-practice violations.

- **PrivacyAssist**: Designed a multi-agent privacy assistant combining on-device/server-side agents, **RAG-enhanced LLM reasoning**, and automatic summarization to generate user-facing warnings; evaluated with **200 participants** and **2,347 apps**.

Industry Engineering Experience

Senior DevOps Engineer 10/2021 – 12/2022
Home Credit Vietnam – OpenAPI Team Ho Chi Minh City, Vietnam

- **OpenAPI Platform / Kong API Gateway**: Designed and maintained CI/CD pipelines for OpenAPI services using **GitLab CI/CD, Jenkins, Harbor Registry, Docker, and Kubernetes**; automated Kong installation, patching, and configuration using **Ansible** and **Python**.
- **Kong API Gateway Automation**: Implemented automation for Kong **routes, services, consumers, ACLs, plugins, and authentication methods**, improving deployment repeatability and reducing manual configuration effort.
- **Zero-Downtime Release and Service Mesh**: Built canary deployment workflows using **Istio service mesh, Kubernetes, and Python scripts** to support safer production releases for microservice systems.
- **Backup, Disaster Recovery, and Security Hardening**: Developed automated Kong backup and **Disaster Recovery Center (DRC)** deployment workflows; collaborated with security teams on **F5 firewall**-based cyber-defense scenarios for API gateway infrastructure.
- **Core Technologies**: **Kubernetes, Docker, Jenkins, GitLab CI/CD, Harbor Registry, Kong API Gateway, Ansible, Python, Istio, GitOps, F5 firewall, Linux**.

Backend Developer and DevOps Engineer 01/2018 – 09/2021
VNPT Information Technology Company (VNPT IT) Ho Chi Minh City, Vietnam

- **VNPT One Login – Single Sign-On Platform**: Worked as **Project Manager** and backend developer; designed the system architecture and developed backend APIs using **Go**; researched and integrated **CAS Apereo** and **OAuth** for authentication and identity management.
- **VNPT E-Conference – Online Conferencing Platform**: Worked as **Project Manager**; researched, designed, and deployed a production conferencing platform based on **BigBlueButton**; analyzed modular components including **Kurento, FreeSWITCH, database services, and backend modules**.
- **VNPT ORIMX, Open Data, GIS Platform, Drive, and SMS Brandname**: Worked as **DevOps Engineer** for multiple backend and platform services; deployed services through CI/CD pipelines using **Jenkins, Docker, Kubernetes, and Rancher**.
- **Core Technologies**: **Go, CAS Apereo, OAuth, BigBlueButton, Kurento, FreeSWITCH, Jenkins, Docker, Kubernetes, Rancher, Linux, CI/CD, backend APIs**.

Software Engineer 12/2016 – 12/2017
Tecapro Telecom Ho Chi Minh City, Vietnam

- **Asterisk PBX / VoIP Management System**: Researched and developed telecommunication systems based on **Asterisk PBX**; built an **Asterisk Management Interface (AMI)** integration using **PHP**; developed a web-based application for real-time monitoring and control of PBX functionalities.
- **Core Technologies**: **Asterisk PBX, AMI, PHP, VoIP, web-based monitoring systems**.

Telecommunications Engineer 05/2015 – 11/2016
Saigontourist Cable Television (SCTV) Ho Chi Minh City, Vietnam

- **Telecommunication and Digital-Service R&D**: Researched and deployed **GPON network infrastructure**, **IPTV systems**, and digital-service solutions; collaborated with **Samsung Research** on **Tizen OS applications** and set-top box solutions.
- **Payment Gateway and Smart Home Solutions**: Designed and implemented a payment-gateway solution for online transactions; collaborated with **Lumi Vietnam** to research and develop Smart Home solutions.
- **Core Technologies**: **GPON, IPTV, Tizen OS, set-top box systems, payment gateway, Smart Home, telecommunications infrastructure**.

Education

Ph.D. in Computer Science 01/11/2022 – 11/2025
University of Insubria Varese, Italy
Field of study: Security and Privacy.
Thesis title: *Security and Privacy Analysis of Mobile and Wearable Applications*.

Master of Engineering in Telecommunications Engineering 01/09/2015 – 26/04/2018

University of Technology – Vietnam National University Ho Chi Minh City Ho Chi Minh City, Vietnam
Field of study: Telecommunications.
Thesis title: *Elliptic Curve Cryptography algorithm and application to build a Smart-Auto Parking system.*
Thesis grade: 8.2/10.

Bachelor of Engineering in Electrical-Electronics Engineering 01/09/2010 – 15/04/2015
University of Technology – Vietnam National University Ho Chi Minh City Ho Chi Minh City, Vietnam
Honors Program, Electronics-Telecommunications.
Field of study: Telecommunications.
Thesis title: *Integrated smart parking and RFID-based access-control system.*
Thesis grade: 9.17/10.

Selected Publications

Journal Articles

1. Tran Thanh Lam Nguyen, Barbara Carminati, and Elena Ferrari. *WearPri: Compliance and responsibility estimation in wearable apps*. IEEE Transactions on Dependable and Secure Computing, 2026. Manuscript under review.
2. Tran Thanh Lam Nguyen, Barbara Carminati, and Elena Ferrari. *ALIBIS: Assessing and mitigating the risk of sensitive metadata leakage in mobile image sharing*. Pervasive and Mobile Computing, 117:102171, 2026. doi: 10.1016/j.pmcj.2026.102171.
3. Tran Thanh Lam Nguyen, Son Xuan Ha, Trieu Hai Le, Huong Hoang Luong, Khanh Hong Vo, Khoi Huynh Tuan Nguyen, Tuan Anh Dao, Hy Vuong Khang Nguyen, et al.. *BMDD: A novel approach for IoT platform*. PeerJ Computer Science, 8:e950, 2022. doi: 10.7717/peerj-cs.950.

Conference Proceedings

1. Tran Thanh Lam Nguyen, Edoardo Di Tullio, Barbara Carminati, and Elena Ferrari. *PrivacyAssist: A user-centric agent framework for detecting privacy inconsistencies in Android apps*. Proceedings of the 19th ACM Conference on Security and Privacy in Wireless and Mobile Networks, 2026. Accepted for publication. arXiv: 2604.23248.
2. Tran Thanh Lam Nguyen, Barbara Carminati, and Elena Ferrari. *Detecting privacy non-compliance in wearable apps via knowledge graphs and LLMs*. IEEE WiMob 2025, pp. 1–7, 2025. doi: 10.1109/WiMob66857.2025.11257567.
3. Tran Thanh Lam Nguyen, Barbara Carminati, and Elena Ferrari. *MetaLeak: Assessing image metadata leakage in Android apps*. IEEE/ACS AICCSA 2024, pp. 1–10, 2024. doi: 10.1109/AICCSA63423.2024.10912621.
4. Tran Thanh Lam Nguyen, The Anh Nguyen, Hong Khanh Vo, Hoang Huong Luong, Khoi Nguyen Huynh Tuan, Anh Tuan Dao, and Ha Xuan Son. *Toward a security IoT platform with high-rate transmission and low energy consumption*. ICCSA 2021, Springer, pp. 647–662, 2021. doi: 10.1007/978-3-030-86653-2_47.

Book Chapter

1. Tran Thanh Lam Nguyen, Barbara Carminati, and Elena Ferrari. *LLMs on support of privacy and security of mobile apps: State-of-the-art and research directions*. In *AI for Cybersecurity: Research and Practice*, Wiley-IEEE Press, pp. 29–66, 2025. doi: 10.1002/9781394293773.ch02.

Awards

- 2020 – VNPT Creativity Contest, Third Prize, *Building a chatbot system for medical consultation*.
- 2019 – VNPT Creativity Contest, Third Prize, *Building a smart IPS system for patient monitoring*.

Referees

Prof. Elena Ferrari
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